

Total No. of Questions : 4]

SEAT No. :

PA-10288

[Total No. of Pages : 1

[6009]-322

T.E. (Computer Engineering) (Insem.)
DATA SCIENCE AND BIG DATA ANALYTICS
(2019 Pattern) (Semester - II) (310251)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer questions Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific calculator is allowed.

- Q1)** a) What are dimensionality reduction and its benefits? [4]
b) What is data wrangling? Why do you need it? [5]
c) What is regression? Explain different types of regression with example. [6]

OR

- Q2)** a) Differentiate between Data Science, Machine Learning and AI. [4]
b) What does feature engineering typically includes? [5]
c) What is Data Discretization, explain Forms of data discretization. [6]
- Q3)** a) Write a short note on contingency table, explain with example. [4]
b) With an example explain Baye's theorem. Also explain its key terms. [5]
c) Is there a correlation between the variables in the following data set? [6]

Hours	9	15	25	14	10	18	19	16	20	18
Marks	39	56	93	61	50	75	42	70	66	32

OR

- Q4)** a) What is population & how is it differ from a sample? [4]
b) With an example, explain one-tailed & two-tailed t-tests. [5]
c) Describe the Chi-Square Test of Independence. [6]



Total No. of Questions : 4]

SEAT No. :

PB-103

[Total No. of Pages : 2

[6269]-317

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(2019 Pattern) (Semester - II) (310251)

Time : 1 Hour]

[Max. Marks : 30

Instructions to the candidates:

- 1) Answer Q.1 or Q.2, Q.3 or Q.4.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right indicate full marks.
- 4) Assume suitable data if necessary.
- 5) Use of Scientific Calculator is permitted.

- Q1)** a) Explain data wrangling methods with suitable example. [5]
b) Suppose that the data for analysis includes the attribute age, given the following data (in increasing order) for the attribute age: 13, 15, 16, 16, 19, 20, 20, 21, 22, 22, 25, 25, 25, 25, 30, 33, 33, 35, 35, 35, 35, 36, 40, 45, 46, 52, 70. [5]
i) Use smoothing by bin means, using a bin depth of 3.
ii) What other methods are there for data smoothing?
c) What is data science? Compare data science and information science. [5]

OR

- Q2)** a) Explain 5 V's of Big Data. [5]
b) Explain different phases of data analytics life cycle with neat diagram. [5]
c) Compare Business Intelligence and data science. [5]
- Q3)** a) Explain skewness and kurtosis. What is the purpose of finding skewness of data? [5]
b) What is degree of freedom? Explain with example. [5]
c) How hypothesis testing works? Explain steps. [5]

OR

P.T.O.

- Q4)** a) List out measures of dispersion with their significance and mathematical formulae. [5]
- b) Describe Chi-square Goodness of Fit test. [5]
- c) Assume that a patient X took a lab test for a certain disease and tested positive. The lab test returns a positive result in 95% of the cases in which the disease is actually present and it falsely returns a positive result in 6% of the cases in which the disease is not present. Further more only 1% of the entire population has this disease. What is the probability that X actually has the disease given that he is tested positive. [5]

